TAGORE COLLEGE OF ARTS AND SCIENCE

(Affiliated to University of Madras)
Chrompet, Chennai-44



DEPARTMENT OF COMPUTER SCIENCE

RELATIONAL DATABASE MANAGEMENT SYSTEM

NAME :

CLASS :

REG NO :

TAGORE COLLEGE OF ARTS AND SCIENCE

(Affiliated to the University of Madras)
Chrompet, Chennai – 600 044

DEPARTMENT OF COMPUTER SCIENCE

Certified to be the Bonafide Record of Practical work done by Mr. /Miss. ……. Reg. No….....of Third Year B.Sc., in the Department of Computer Science, Shift-I, Tagore College of Arts & Science, Chrompet, Chennai – 600 044 during the year 2026-2027

LECTURER IN – CHARGE

HEAD OF THE DEPARTMENT

Submitted for B.Sc., Degree Practical Examination held on ……

at the Department of Computer Science, Shift-I, Tagore College of Arts and Science, Chrompet, Chennai –44.

Date :

Examiners

1.

2.

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1)DDL COMMENTS WITH CONSTRAINTS

DDL commands:

- ✴ CREATE
- ✴ ALTER
- ✴ DROP
- ✴ RENAME
- ✴ TRUNCATE

Queries:

CREATE TABLE

Q1 → create table student(stid number, stname varchar(20), score integer);

Q2 → desc student;

Q3 → create table stu(sid number primary key, sname varchar(20) not null, email varchar(30) unique, age decimal(10,2) check (age>=18));

Q4 → desc stu;

ALTER TABLE

Q5 → alter table student drop column score;

Q6 → alter table student add gender varchar(5);

Q7 → alter table stu modify age number;

RENAME TABLE

Q8 → rename student to stu10;

Q9 → desc stu10;

Q10 → alter table stu rename column email to emailid;

Q11 → desc stu;

TRUNCATE AND DROP

Q12 → truncate table stu;

Q13 → drop table stu;

Q14 → desc stu;

OUTPUT:

CREATE TABLE

O/P → Table created.

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STUDENT	STID	Number	-	-	-	-	✓	-	-
	STNAME	Varchar2	20	-	-	-	✓	-	-
	SCORE	Number	-	-	0	-	✓	-	-

O/P →

O/P → Table created.

O/P →

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STU	SID	Number	-	-	-	1	-	-	-
	SNAME	Varchar2	20	-	-	-	-	-	-
	EMAIL	Varchar2	30	-	-	-	✓	-	-
	AGE	Number	-	10	2	-	✓	-	-

ALTER TABLE

O/P →

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STUD	STID	Numbe	-	-	-	-	✓	-	-
	STN	Varcha	20	-	-	-	✓	-	-

O/P → Table altered.

O/P → Table altered.

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STU	SID	Number	-	-	-	1	-	-	-
	SNAME	Varchar2	20	-	-	-	-	-	-
	EMAIL	Varchar2	30	-	-	-	✓	-	-
	AGE	Number	-	-	-	-	✓	-	-

RENAME TABLE

O/P → Statement processed.

O/P →

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STU10	STID	Number	-	-	-	-	✓	-	-
	STNAME	Varchar2	20	-	-	-	✓	-	-
	GENDER	Varchar2	5	-	-	-	✓	-	-

O/P →

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
STU	SID	Number	-	-	-	1	-	-	-
	SNAME	Varchar2	20	-	-	-	-	-	-
	EMAILID	Varchar2	30	-	-	-	✓	-	-
	AGE	Number	-	-	-	-	✓	-	-

TRUNCATE AND DROP

O/P → Table truncated.

O/P → Table dropped.

O/P → Object to be described could not be found.

2)DML COMMENTS WITH CONSTRAINTS

DML commands:

- ✴ INSERT
- ✴ UPDATE
- ✴ DELETE
- ✴ SELECT

Queries:

INSERT QUERY

Q1 → insert into stu10(stid,stname,gender) values (1001,'aaa','f');

Q2 → insert into stu10(stid,stname,gender) values (1002,'bbb','f');

Q3 → insert into stu10(stid,stname,gender) values (1003,'cccc','f');

Q4 → insert into stu10(stid,stname,gender) values (1004,'ddd','m');

Q5 → insert into stu10(stid,stname,gender) values (1005,'EEE','M');

SELECT QUERY

Q6 → select * from stu10;

Q7 → select stname from stu10;

Q8 → select * from stu10 where gender='f';

UPDATE QUERY

Q9 → update stu10 set stname='anu' where stid=1001;

Q10 → update stu10 set stname='bala', gender='m' where stid=1002;

Q11 → select * from stu10;

DELETE QUERY

Q12 → delete from stu10 where stid=1003;

Q13 → select * from stu10;

Q14 → delete from stu10;

Q15 → select * from stu10;

OUTPUT:

INSERT QUERY

O/P → 1 row(s) inserted.

SELECT QUERY

O/P
→

STID	STNAME	GENDER
1001	aaa	f
1002	bbb	f
1003	cccc	f
1004	ddd	m
1005	EEE	M

O/P →

STNAME
aaa
bbb
cccc
ddd
EEE

O/P →

STID	STNAME	GENDER
1001	aaa	f
1002	bbb	f
1003	cccc	f

UPDATE QUERY

O/P → 1 row(s) updated.

O/P → 1 row(s) updated.

O/P →

STID	STNAME	GENDER
1001	anu	f
1002	bala	m
1003	cccc	f
1004	ddd	m
1005	EEE	M

DELETE QUERY

O/P → 1 row(s) deleted.

O/P →

STID	STNAME	GENDER
1001	anu	f
1002	bala	m
1004	ddd	m
1005	EEE	M

O/P → 4 row(s) deleted.

O/P → no data found

3)QUERIES, SUBQUERIES AND AGGREGATE FUNCTION

Queries:

BASIC QUERY

Q1 → create table empl(eid number primary key, ename varchar(20), deptid number, deptname varchar(20),salary number);

Q2 → insert into empl(eid,ename,deptid,deptname,salary) values (1201,'Arun',104,'CS',50000);

Q3 → insert into empl(eid,ename,deptid,deptname,salary) values (1202,'Bala',104,'CS',60000);

Q4 →insert into empl(eid,ename,deptid,deptname,salary) values (1203,'Chandru',105,'COMMERCE',40000);

Q5 →insert into empl(eid,ename,deptid,deptname,salary) values (1204,'Dilip',104,'CS',45000);

Q6 →insert into empl(eid,ename,deptid,deptname,salary) values (1205,'Dhanu',106,'TAMIL',30000);

Q7 →insert into empl(eid,ename,deptid,deptname,salary) values (1206,'Guru',106,'TAMIL',58000);

Q8 →insert into empl(eid,ename,deptid,deptname,salary) values (1207,'Hari',105,'COMMERCE',38000);

SELECT QUERY

Q9 → select * from empl;

Q10 → select eid,ename,salary from empl;

Q11 → select * from empl where salary>=50000;

AGGREGATE FUNCTION

Q12 → select COUNT(*) as totalemp from empl;

Q13 → select SUM(salary) as totalsal from empl;

Q14 → select AVG(salary) as totalsal from empl;

Q15 →select MAX(salary) as maxsalary from empl;

Q16 →select MIN(salary) as minsalary from empl;

Q17 →select eid,salary from empl order by salary desc;

Q18 →select deptid, sum(salary)as sal from empl group by deptid;

SUBQUERIES

Q19 → select * from empl where salary > (select avg(salary) from empl);

Q20 →create table dept(deptid number,deptname varchar(20));

Q21 →insert into dept(deptid,deptname) values (104,'CS');

Q22 →insert into dept(deptid,deptname) values (105,'COMMERCE');

Q23 →insert into dept(deptid,deptname) values (106,'TAMIL');

Q24 →insert into dept(deptid,deptname) values (107,'ENGLISH');

Q25 →select * from dept;

--- List department wise employee count:

Q26 →select d.deptname, e.empcount from dept d JOIN (select deptid, count(*) as empcount from empl group by deptid) e on d.deptid = e.deptid;

JOIN QUERIES

Q27 →select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e join dept d on e.deptid = d.deptid;

Q28 → select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e LEFT JOIN dept d on e.deptid = d.deptid;

Q29 → select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e RIGHT JOIN dept d on e.deptid = d.deptid;

Q30 → select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e CROSS JOIN dept d;

OUTPUT:

BASIC QUERY

O/P → Table created.

SELECT QUERY

O/P
→

EID	ENAME	DEPTID	DEPTNAME	SALARY
1201	Arun	104	CS	50000
1202	Bala	104	CS	60000
1203	Chandru	105	COMMERCE	40000
1204	Dilip	104	CS	45000
1205	Dhanu	106	TAMIL	30000
1206	Guru	106	TAMIL	58000
1207	Hari	105	COMMERCE	38000

O/P →

EID	ENAME	SALARY
1201	Arun	50000
1202	Bala	60000
1203	Chandru	40000
1204	Dilip	45000
1205	Dhanu	30000
1206	Guru	58000
1207	Hari	38000

O/P →

EID	ENAME	DEPTID	DEPTNAME	SALARY
1201	Arun	104	CS	50000
1202	Bala	104	CS	60000
1206	Guru	106	TAMIL	58000

AGGREGATE FUNCTION

O/P →

TOTALEMP
7

O/P →

TOTALSAL
321000

O/P →

TOTALSAL
45857.142857142857142857

O/P
→

MINSALARY
30000

O/P

→EID	SALARY
1202	60000
1206	58000
1201	50000
1204	45000
1203	40000
1207	38000
1205	30000

O/P →

DEPTID	SAL
104	155000
105	78000
106	88000

SUBQUERIES

O/P →

EID	ENAME	DEPTID	DEPTNAME	SALARY
1201	Arun	104	CS	50000

1202	Bala	104	CS	60000
1206	Guru	106	TAMIL	58000

O/P →

DEPTID	DEPTNAME
104	CS
105	COMMERCE
106	TAMIL
107	ENGLISH

O/P →

DEPTNAME	EMPCOUNT
COMMERCE	2
CS	3
TAMIL	2

JOIN QUERIES

O/P →

EID	ENAME	SALARY	DEPTID	DEPTNAME
1201	Arun	50000	104	CS
1202	Bala	60000	104	CS
1203	Chandru	40000	105	COMMERCE
1204	Dilip	45000	104	CS
1205	Dhanu	30000	106	TAMIL
1206	Guru	58000	106	TAMIL

1207	Hari	38000	105	COMMERCE

O/P →

EID	ENAME	SALARY	DEPTID	DEPTNAME
1204	Dilip	45000	104	CS
1202	Bala	60000	104	CS
1201	Arun	50000	104	CS
1207	Hari	38000	105	COMMERCE
1203	Chandru	40000	105	COMMERCE
1206	Guru	58000	106	TAMIL
1205	Dhanu	30000	106	TAMIL

O/P →

EID	ENAME	SALARY	DEPTID	DEPTNAME
1201	Arun	50000	104	CS
1202	Bala	60000	104	CS
1203	Chandru	40000	105	COMMERCE
1204	Dilip	45000	104	CS
1205	Dhanu	30000	106	TAMIL
1206	Guru	58000	106	TAMIL
1207	Hari	38000	105	COMMERCE
-	-	-	107	ENGLISH

O/P →

EID	ENAME	SALARY	DEPTID	DEPTNAME
1201	Arun	50000	104	CS
1202	Bala	60000	104	CS
1203	Chandru	40000	104	CS
1204	Dilip	45000	104	CS
1205	Dhanu	30000	104	CS
1206	Guru	58000	104	CS
1207	Hari	38000	104	CS
1201	Arun	50000	105	COMMERCE
1202	Bala	60000	105	COMMERCE
1203	Chandru	40000	105	COMMERCE

4)PL/SQL – EXCEPTION HANDLING

Queries:

Predefined Exceptions

Q1 →

```
DECLARE
 num1 NUMBER := 10;
 num2 NUMBER := 0;
 result NUMBER;
BEGIN
 result := num1 / num2;
DBMS_OUTPUT.PUT_LINE('Result: ' || result);
EXCEPTION
 WHEN ZERO_DIVIDE THEN
 DBMS_OUTPUT.PUT_LINE('Cannot divide by zero.');
END;
```

User-Defined Exceptions

Q2 →

```
DECLARE
 insufficient_balance EXCEPTION;
 balance NUMBER := 500;
BEGIN
 IF balance < 1000 THEN
 RAISE insufficient_balance;
 END IF;
EXCEPTION
 WHEN insufficient_balance THEN
```

```
 DBMS_OUTPUT.PUT_LINE('Balance is too low.');
END;
```

Named System Exceptions

Q3 →

```
DECLARE
 ename emp.name%TYPE;
BEGIN
 SELECT name into ename
 FROM emp
 WHERE eid = 1001; -- assuming invalid ID
DBMS_OUTPUT.PUT_LINE('Employee Name: ' || ename);
EXCEPTION
 WHEN NO_DATA_FOUND THEN
 DBMS_OUTPUT.PUT_LINE('No employee found with that ID.');
END;
```

Unhandled (OTHERS) Exception

Q4 →

```
BEGIN
 EXECUTE IMMEDIATE 'DROP TABLE nonexistent_table';
EXCEPTION
 WHEN OTHERS THEN
 DBMS_OUTPUT.PUT_LINE('Some unexpected error occurred: ' || SQLERRM);
END;
```

OUTPUT:

Predefined Exceptions

O/P

→

Cannot divide by zero.

User-Defined Exceptions

O/P

→

Balance is too low.

Named System Exceptions

O/P

→

No employee found with that ID.

Unhandled (OTHERS) Exception

O/P

→

Some unexpected error occurred: ORA-00942: table or view does not exist

5)PL/SQL – CURSOR

Queries:

Q1 → select * from emp;

Implicit Cursor

Q2 →

```
BEGIN
 UPDATE emp SET salary = salary + 1000 WHERE eid = 103;

 IF SQL%ROWCOUNT = 1 THEN
 DBMS_OUTPUT.PUT_LINE('1 row updated successfully.');
 END IF;
END;
```

Q3 → select * from emp;

Explicit Cursor

Q4 →

```
DECLARE
 CURSOR emp_cursor IS SELECT eid, name FROM emp;
 v_id emp.eid%TYPE;
 v_name emp.name%TYPE;
BEGIN
 OPEN emp_cursor;
 LOOP
 FETCH emp_cursor INTO v_id, v_name;
 EXIT WHEN emp_cursor%NOTFOUND;
 DBMS_OUTPUT.PUT_LINE('ID: ' || v_id || ', Name: ' || v_name);
```

```
 END LOOP;
 CLOSE emp_cursor;
END;
```

Cursor FOR Loop

Q5 →

```
BEGIN
 FOR emp_rec IN (SELECT eid, name FROM emp) LOOP
 DBMS_OUTPUT.PUT_LINE('ID: ' || emp_rec.eid || ', Name: ' || emp_rec.name);
 END LOOP;
END;
```

Parameterized Cursor

Q6 →

```
DECLARE
 CURSOR emp_cursor(p_dept_id NUMBER) IS
 SELECT eid, name FROM emp WHERE dept_id = p_dept_id;
 v_id emp.eid%TYPE;
 v_name emp.name%TYPE;
BEGIN
 OPEN emp_cursor(123); -- passing department id
 LOOP
 FETCH emp_cursor INTO v_id, v_name;
 EXIT WHEN emp_cursor%NOTFOUND;
 DBMS_OUTPUT.PUT_LINE('ID: ' || v_id || ', Name: ' || v_name);
 END LOOP;
 CLOSE emp_cursor;
END;
```

OUTPUT:

O/P

→

EID	NAME	DOJ	SALARY	DEPT_ID
103	Cholan	12-FEB-20	30000	123
104	Hari	22-FEB-24	20000	122
101	Arun	12-FEB-20	40000	123
102	Bala	01-DEC-19	50000	121
105	Guru	10-JAN-18	60000	122
106	Mathi	01-JUN-20	46000	121

Implicit Cursor

O/P →

1 row updated successfully.

1 row(s) updated.

O/P →

EID	NAME	DOJ	SALARY	DEPT_ID
103	Cholan	12-FEB-20	31000	123
104	Hari	22-FEB-24	20000	122
101	Arun	12-FEB-20	40000	123
102	Bala	01-DEC-19	50000	121
105	Guru	10-JAN-18	60000	122
106	Mathi	01-JUN-20	46000	121

Explicit Cursor

O/P

→

ID: 103, Name: Cholan

ID: 104, Name: Hari

ID: 101, Name: Arun

ID: 102, Name: Bala

ID: 105, Name: Guru

ID: 106, Name: Mathi

Cursor FOR Loop

O/P

→

ID: 103, Name: Cholan

ID: 104, Name: Hari

ID: 101, Name: Arun

ID: 102, Name: Bala

ID: 105, Name: Guru

ID: 106, Name: Mathi

Parameterized Cursor

O/P

→

ID: 103, Name: Cholan

ID: 101, Name: Arun

6)PL/SQL – TRIGGER

PL/SQL Queries:

Q1 → CREATE TABLE students (roll_no NUMBER PRIMARY KEY,
name VARCHAR2(50), marks NUMBER);

BEFORE INSERT TRIGGER

Q2 →

```
CREATE OR REPLACE TRIGGER trg_before_insert_students
BEFORE INSERT ON students
FOR EACH ROW
BEGIN
 -- If marks not given, set to 0
 IF :NEW.marks IS NULL THEN
 :NEW.marks := 0;
 END IF;
END;
```

AFTER INSERT TRIGGER

Q3 →

```
CREATE OR REPLACE TRIGGER trg_after_insert_students
AFTER INSERT ON students
BEGIN
 DBMS_OUTPUT.PUT_LINE('A new student was added!');
END;
```

Q4 → INSERT INTO students (roll_no, name) VALUES (1, 'John');

Q5 → select * from students;

BEFORE UPDATE TRIGGER

Q6 →

```
CREATE OR REPLACE TRIGGER trg_before_update_students
BEFORE UPDATE ON students
FOR EACH ROW
BEGIN
 -- Prevent marks from being more than 100
 IF :NEW.marks > 100 THEN
 :NEW.marks := 100;
 END IF;
END;
```

Q7 → UPDATE students SET marks = 120 WHERE roll_no = 1;

Q8 → select * from students;

AFTER DELETE TRIGGER

Q9 →

```
CREATE OR REPLACE TRIGGER trg_after_delete_students
AFTER DELETE ON students
BEGIN
 DBMS_OUTPUT.PUT_LINE('A student record was deleted!');
END;
```

Q10 → DELETE FROM students WHERE roll_no = 1;

OUTPUT:

O/P →

Table Created.

BEFORE INSERT TRIGGER

O/P →

Trigger created.

AFTER INSERT TRIGGER

O/P →

Trigger created.

O/P →

A new student was added!

1 row(s) inserted.

O/P
→

ROLL_NO	NAME	MARKS
1	John	0

BEFORE UPDATE TRIGGER

O/P →

Trigger created.

O/P →

1 row(s) updated.

O/P
→

ROLL_NO	NAME	MARKS
1	John	100

AFTER DELETE TRIGGER

O/P → Trigger created.

O/P → A student record was deleted!
1 row(s) deleted.

7)PL/SQL – PACKAGES

.

PL/SQL Queries:

Package Specification

Q1 →

```
CREATE OR REPLACE PACKAGE math_pkg AS
        FUNCTION square_num(p_num NUMBER) RETURN NUMBER;
END math_pkg;
```

Package Body

Q2 →

```
CREATE OR REPLACE PACKAGE BODY math_pkg AS
 FUNCTION square_num(p_num NUMBER) RETURN NUMBER IS
 BEGIN
 RETURN p_num * p_num;
 END;
END math_pkg;
```

Test the Package

Q3 →

```
BEGIN
 DBMS_OUTPUT.PUT_LINE('Square of 5: ' || math_pkg.square_num(5));
END;
```

Predefined Package

Q4 →

```
BEGIN
 -- Random number between 1 and 50
 DBMS_OUTPUT.PUT_LINE ('Random Number: ' || ROUND(DBMS_RANDOM.VALUE(1, 50)));

 -- Random uppercase string of length 8
 DBMS_OUTPUT.PUT_LINE('Random String: ' || DBMS_RANDOM.STRING('U', 8));
END;
```

OUTPUT:

Package Specification

O/P →

Package created.

Package Body

O/P →

Package Body created.

Test the Package

O/P →

Square of 5: 25

Predefined Package

O/P

→

Random Number: 34

Random String: ZOWXBLHK

8)LIBRARY MANAGEMENT SYSTEM

Coding:

SQL:

```
create table book ( bookid number primary key,
        bookname varchar2(30) ,
        author varchar2(20) ,
        quan number
        );
```

Form1 (VB 2010):

```
Imports System.Data.OleDb

Public Class Form1
   ' Form-level connection to Oracle 10g
   Dim cn As New OleDbConnection("Provider=MSDAORA;Data Source=XE;User
ID=system;Password=system;")

   ' --- 1. ADD BOOK ---
   Private Sub Button1_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button1.Click
      ' 1. Check if any of your textboxes are empty first
      If TextBox1.Text = "" Or TextBox2.Text = "" Or TextBox3.Text = "" Or TextBox4.Text = "" Then
         MsgBox("Please fill in all fields before adding a book.")
         Exit Sub ' Stops the code here so Oracle doesn't crash
      End If

      Try
         cn.Open()
         Dim sql As String = "INSERT INTO book VALUES (" & TextBox1.Text & ", '" & TextBox2.Text
                  & "', '" & TextBox3.Text & "', " & TextBox4.Text & ")"
         Dim cmd As New OleDbCommand(sql, cn)
         cmd.ExecuteNonQuery()
         MsgBox("Book Added Successfully!")
         TextBox1.Clear()
         TextBox2.Clear()
         TextBox3.Clear()
         TextBox4.Clear()
      Catch ex As Exception
         MsgBox("Error adding book: " & ex.Message)
      Finally
         cn.Close()
      End Try
   End Sub
```

```
' --- 2. ISSUE BOOK ---
Private Sub Button2_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button2.Click
    ' We use InputBox here so you don't have to fill out the textboxes just to issue a book
    Dim bid As String = InputBox("Enter the Book ID to Issue:")
    If bid = "" Then Exit Sub

    Try
        cn.Open()
        Dim sql As String = "UPDATE book SET quan = quan - 1 WHERE bookid = " & bid
        Dim cmd As New OleDbCommand(sql, cn)
        Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

        If rowsAffected > 0 Then
            MsgBox("Book Issued Successfully!")
        Else
            MsgBox("No record found with ID: " & bid)
        End If
    Catch ex As Exception
        MsgBox("Error issuing book: " & ex.Message)
    Finally
        cn.Close()
    End Try
End Sub

' --- 3. RETURN BOOK ---
Private Sub Button3_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button3.Click
    Dim bid As String = InputBox("Enter the Book ID to Return:")
    If bid = "" Then Exit Sub

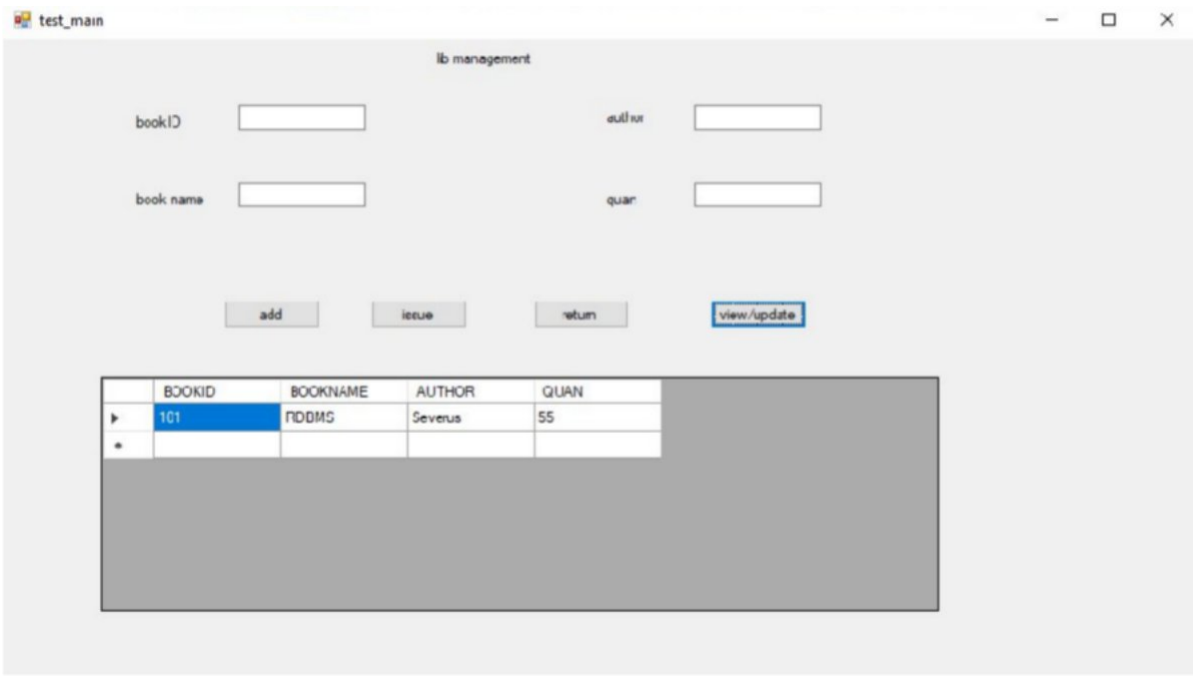
    Try
        cn.Open()
        Dim sql As String = "UPDATE book SET quan = quan + 1 WHERE bookid = " & bid
        Dim cmd As New OleDbCommand(sql, cn)
        Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

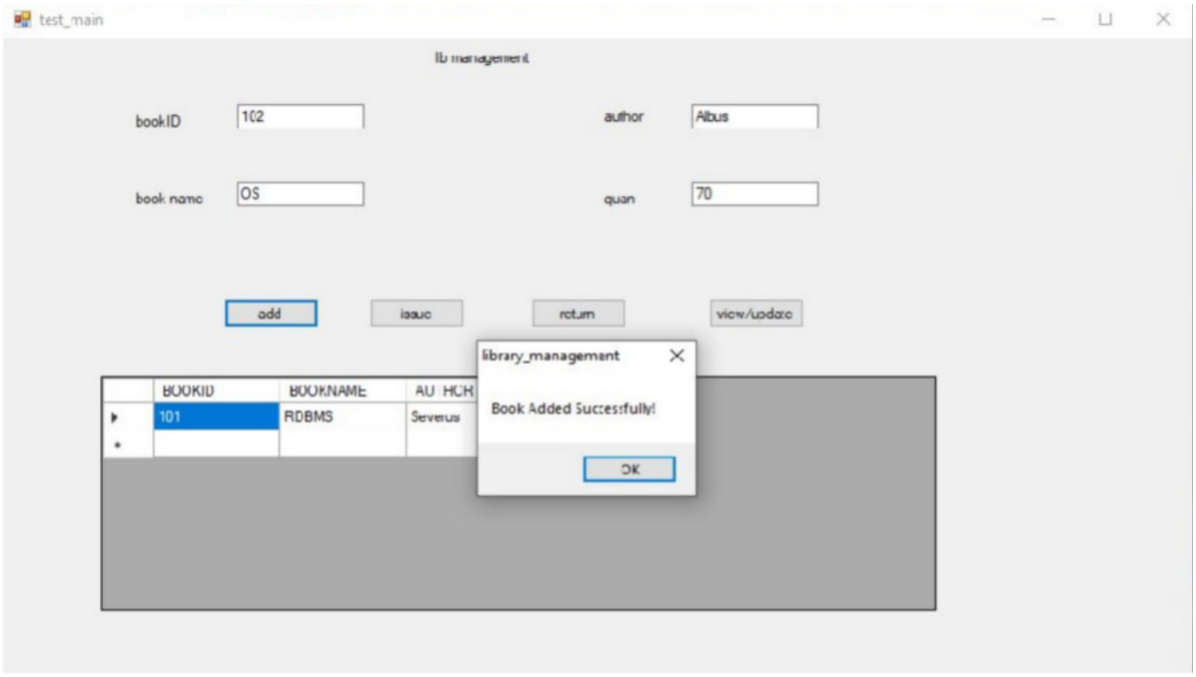
        If rowsAffected > 0 Then
            MsgBox("Book Returned Successfully!")
        Else
            MsgBox("No record found with ID: " & bid)
        End If
    Catch ex As Exception
        MsgBox("Error returning book: " & ex.Message)
    Finally
        cn.Close()
    End Try
End Sub
```

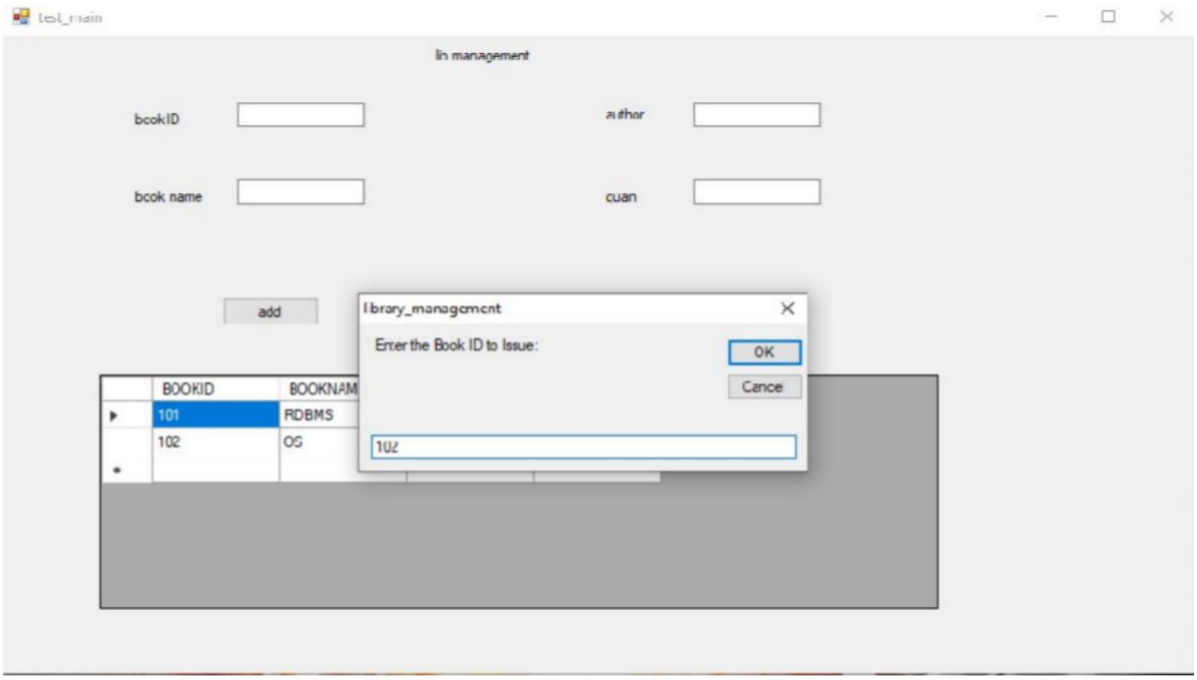
```
    ' --- 4. VIEW REPORT (Updates the DataGridView at the bottom) ---
    Private Sub Button4_Click(ByVal sender As Object, ByVal e As EventArgs) Handles Button4.Click
        Try
            Dim sql As String = "SELECT * FROM book"
            ' DataAdapter grabs the data without needing to manually open/close the connection
            Dim da As New OleDbDataAdapter(sql, cn)
            Dim dt As New DataTable()

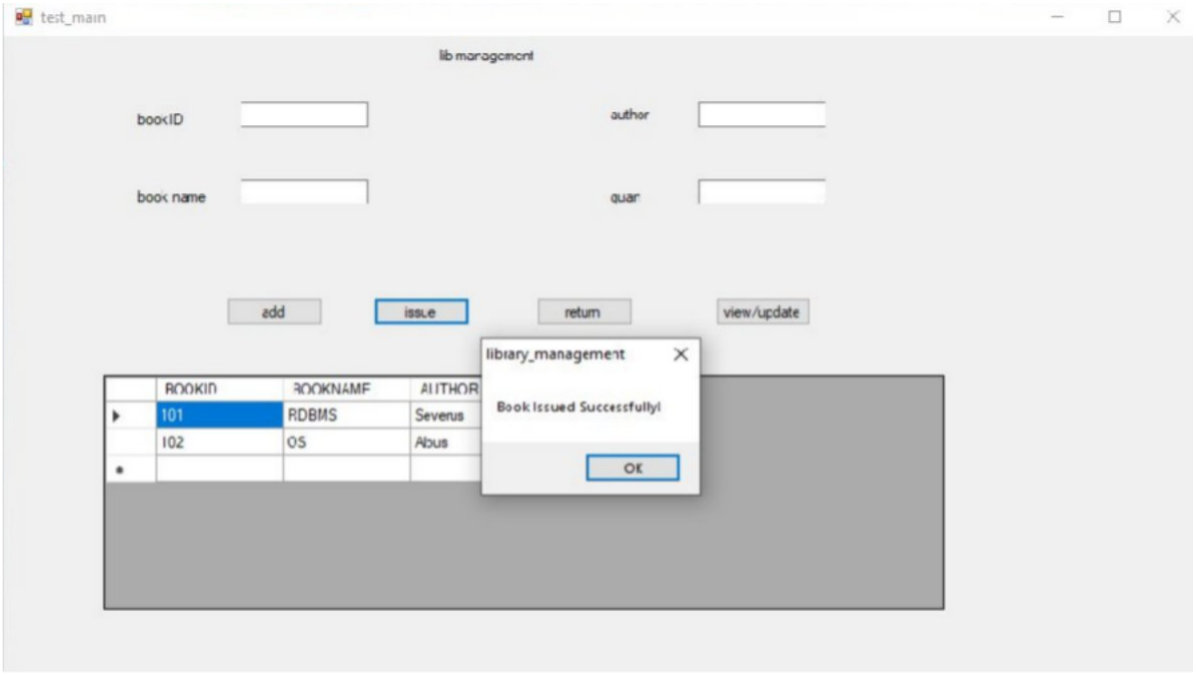
            da.Fill(dt)
            DataGridView1.DataSource = dt
        Catch ex As Exception
            MsgBox("Error loading data: " & ex.Message)
        End Try
    End Sub
End Class
```

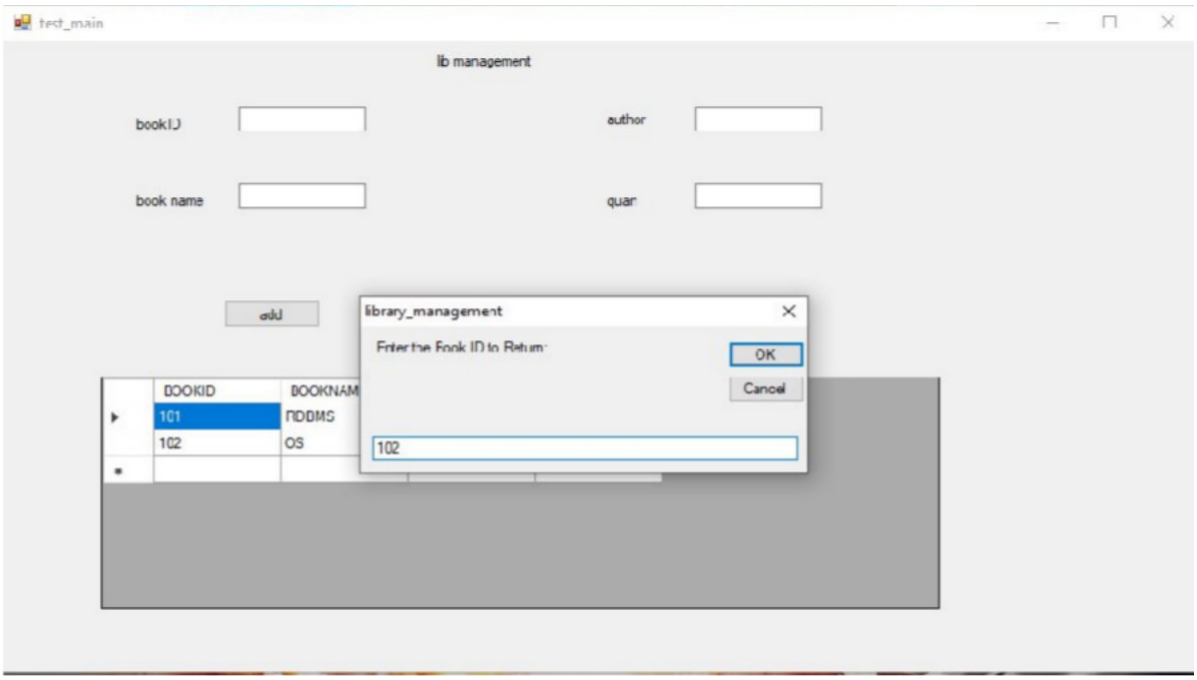
Output:

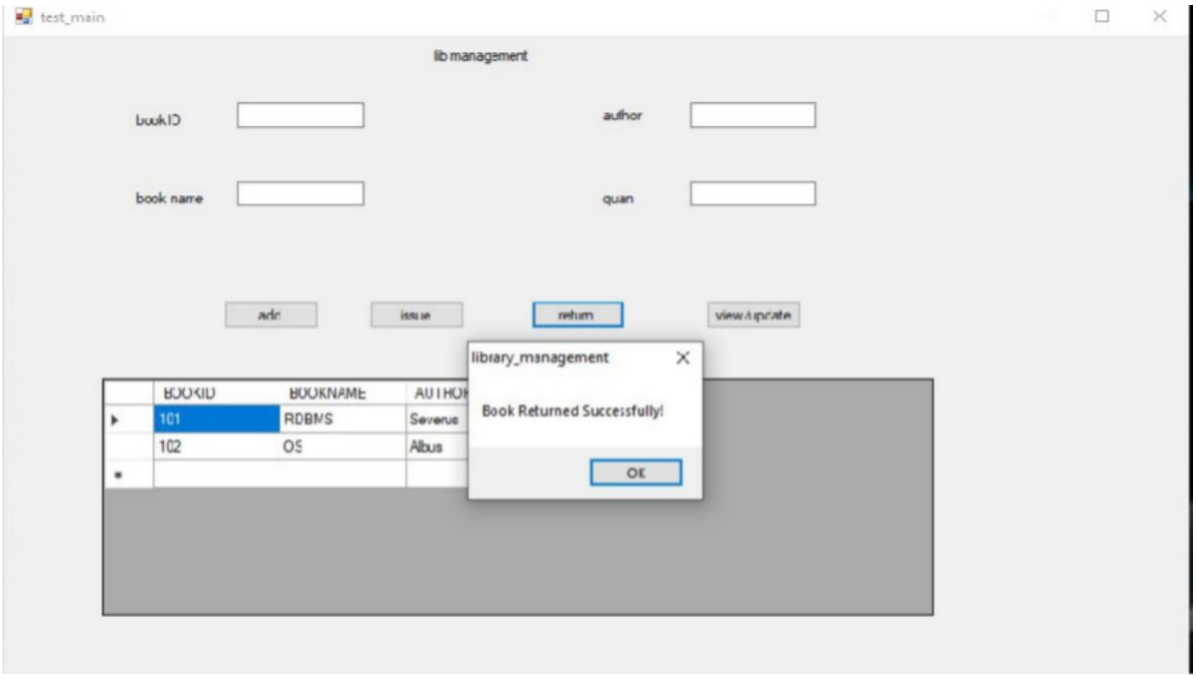

test_main
lib management
bookID
book name
add
issue
return
view/update
BOOKID
BOOKNAME
AUTHOR
QUAN
101
RDBMS
Severus
55


test_main
bookID
102
book name
OS
author
Albus
quan
70
add
BOOKID
BOOKNAME
101
RDBMS
Severus
library_management
Book Added Successfully!
OK


lib management
bcokID
author
bcok name
cuan
add
library_management
Enter the Book ID to Issue:
OK
Cance
102
BOOKID
BOOKNAM
101
RDBMS
102
OS


test_main
lib management
bookID
author
book name
quar
add
issue
return
view/update
library_management
Book Issued Successfully!
OK
BOOKID
BOOKNAME
AUTHOR
101
RDBMS
Severus
102
OS
Abus


test_main
lib management
bookID
author
book name
quan
add
library_management
Enter the Book ID to Return
OK
Cancel
102
BOOKID
BOOKNAM
101
RDBMS
102
OS


test_main
lib management
bookID
author
book name
quan
add
issue
return
view/update
library_management
Book Returned Successfully!
OK
BOOKID
BOOKNAME
101
RDBMS
Severus
102
OS
Albus

9)STUDENT MARKSHEET PROCESSING

Coding:

SQL:

```
CREATE TABLE st (
   stid NUMBER PRIMARY KEY,
   stname VARCHAR2(50),
   os NUMBER,
   rdbms NUMBER,
   cn NUMBER,
   se NUMBER,
   total NUMBER,
   result VARCHAR2(10),
   percent NUMBER
);
```

Form1 :

```
Imports System.Data.OleDb

Public Class Form1
   Dim cn As New OleDbConnection("Provider=MSDAORA;Data Source=XE;User
ID=system;Password=system;")

   ' --- CALCULATE ---
   Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
      TextBox7.Text = (Val(TextBox3.Text) + Val(TextBox4.Text) + Val(TextBox5.Text) +
Val(TextBox6.Text)).ToString()

      If Val(TextBox3.Text) <= 30 Or Val(TextBox4.Text) <= 30 Or Val(TextBox5.Text) <= 30
Or Val(TextBox6.Text) <= 30 Then
         TextBox8.Text = "FAIL"
      Else
         TextBox8.Text = "PASS"
      End If

      TextBox9.Text = (Val(TextBox7.Text) / 4).ToString()
   End Sub
```

```
    ' --- INSERT ---
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        Button1.PerformClick()  ' auto-calculate before insert
        Try
            cn.Open()
            Dim sql As String = "INSERT INTO st VALUES (" & TextBox1.Text & ", '" &
TextBox2.Text & "', " & TextBox3.Text & ", " & TextBox4.Text & ", " & TextBox5.Text & ", "
& TextBox6.Text & ", " & TextBox7.Text & ", '" & TextBox8.Text & "', " & TextBox9.Text &
")"
            Dim cmd As New OleDbCommand(sql, cn)
            cmd.ExecuteNonQuery()

            MsgBox("Successfully Inserted")

            TextBox1.Clear(): TextBox2.Clear(): TextBox3.Clear()
            TextBox4.Clear(): TextBox5.Clear(): TextBox6.Clear()
            TextBox7.Clear(): TextBox8.Clear(): TextBox9.Clear()
        Catch ex As Exception
            MsgBox(ex.Message)
        Finally
            cn.Close()
        End Try
    End Sub

    ' --- DELETE ---
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        Dim studentID As String = InputBox("Enter the Student ID to delete:")
        If studentID = "" Then Exit Sub

        Try
            cn.Open()
            Dim cmd As New OleDbCommand("DELETE FROM st WHERE stid = " & studentID,
cn)
            Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

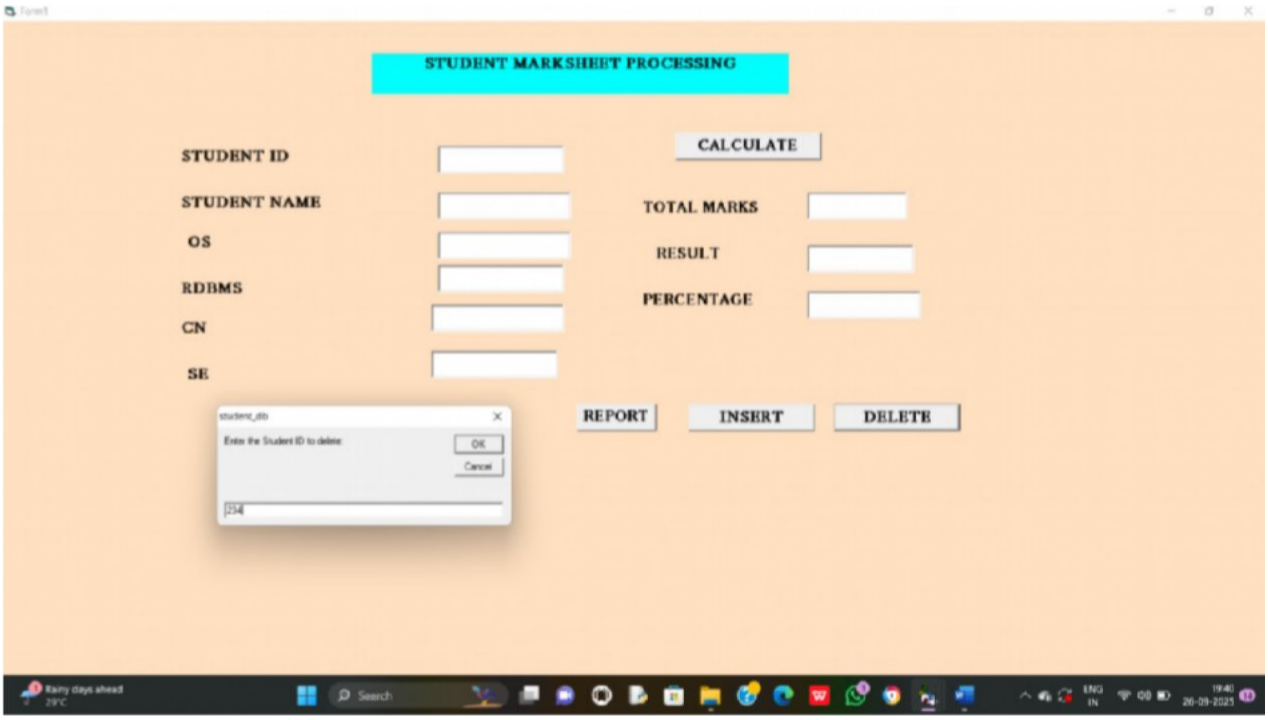
            If rowsAffected > 0 Then
                MsgBox("Record with ID " & studentID & " deleted successfully!")
            Else
                MsgBox("No record found with ID: " & studentID)
            End If
```

```
        Catch ex As Exception
            MsgBox(ex.Message)
        Finally
            cn.Close()
        End Try
    End Sub

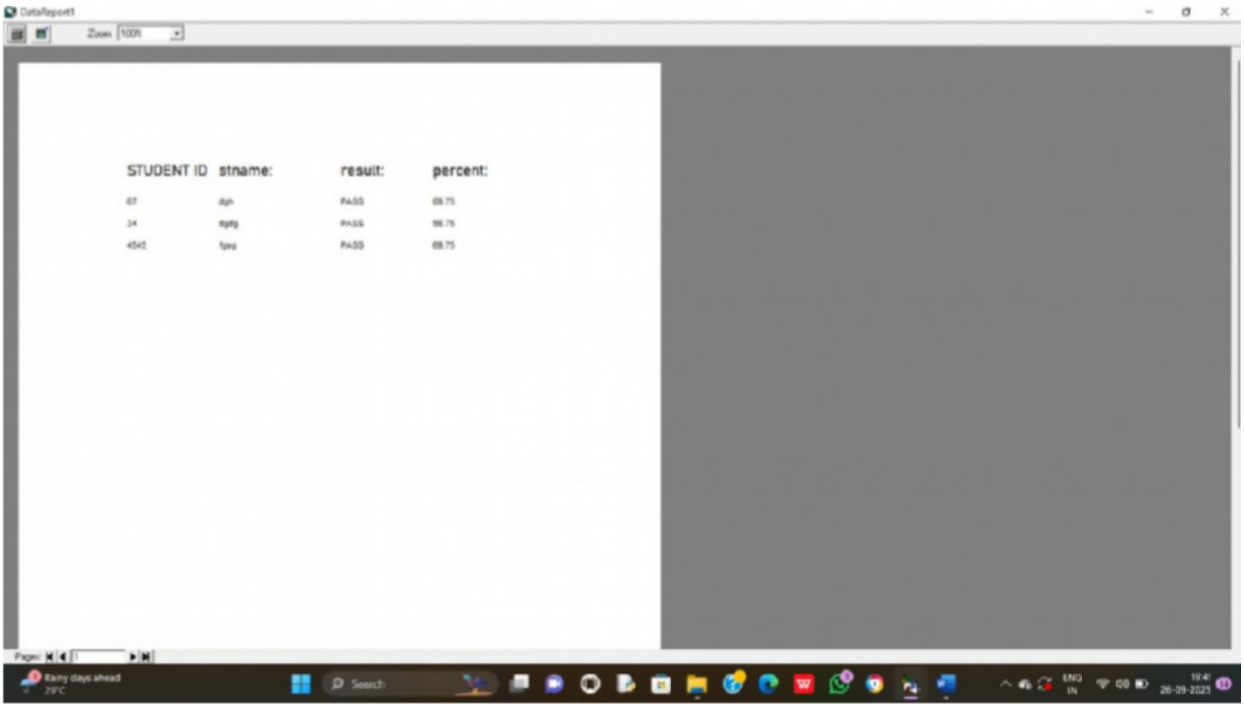
    ' --- VIEW ---
    Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
        Try
            Dim da As New OleDbDataAdapter("SELECT * FROM st", cn)
            Dim dt As New DataTable()
            da.Fill(dt)
            DataGridView1.DataSource = dt
        Catch ex As Exception
            MsgBox(ex.Message)
        End Try
    End Sub
End Class
```

Output:


STUDENT MARKSHEET PROCESSING
STUDENT ID
12345
CALCULATE
STUDENT NAME
Madhu
TOTAL MARKS
324
OS
67
RESULT
PASS
RDBMS
89
PERCENTAGE
81
CN
90
SE
78
REPORT
INSERT
DELETE
student_db
Succesfully Inserted
OK


STUDENT MARKSHEET PROCESSING
STUDENT ID
CALCULATE
STUDENT NAME
TOTAL MARKS
OS
RESULT
RDBMS
PERCENTAGE
CN
SE
REPORT
INSERT
DELETE
student_db
Enter the Student ID to delete
OK
Cancel
234


STUDENT MARKSHEET PROCESSING
STUDENT ID
STUDENT NAME
OS
RDBMS
CN
SE
CALCULATE
TOTAL MARKS
RESULT
PERCENTAGE
REPORT
INSERT
DELETE
student_db
Record with ID 234 deleted successfully!
OK


DataReport1
STUDENT ID
stname:
result:
percent:

10) PAYROLL PROCESSING

Coding:

SQL:

```
CREATE TABLE emp (
   empid   NUMBER PRIMARY KEY,
   empname VARCHAR2(50),
   salary  NUMBER,
   gross   NUMBER,
   net     NUMBER );
```

Form1 :

```
Imports System.Data.OleDb
Public Class Form1
   Dim cn As New OleDbConnection("Provider=MSDAORA;Data Source=XE;User ID=system;
Password=system;") `if data source error occures use orcl instead of XE
   ' --- CALCULATE ---
   Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
      TextBox4.Text = (Val(TextBox3.Text) * 0.25).ToString()                ' Allowance
      TextBox5.Text = (Val(TextBox3.Text) * 0.12).ToString()                ' Deduction
      TextBox6.Text = (Val(TextBox3.Text) + Val(TextBox4.Text)).ToString()     ' Gross
      TextBox7.Text = (Val(TextBox6.Text) - Val(TextBox5.Text)).ToString()     ' Net
   End Sub
   ' --- INSERT ---
   Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
      Button1.PerformClick()  ' auto-calculate before insert
      Try
         cn.Open()
         Dim sql As String = "INSERT INTO emp VALUES (" & TextBox1.Text & ", '" &
TextBox2.Text & "', " & TextBox3.Text & ", " & TextBox6.Text & ", " & TextBox7.Text & ")"
         Dim cmd As New OleDbCommand(sql, cn)
         cmd.ExecuteNonQuery()
         MsgBox("Successfully Inserted")
         TextBox1.Clear(): TextBox2.Clear(): TextBox3.Clear()
         TextBox4.Clear(): TextBox5.Clear(): TextBox6.Clear()
         TextBox7.Clear()
      Catch ex As Exception
         MsgBox(ex.Message)
Finally
         cn.Close()
      End Try
   End Sub
```

```
' --- DELETE ---
  Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    Dim eid As String = InputBox("Enter the EMPLOYEE ID to delete:")
    If eid = "" Then Exit Sub

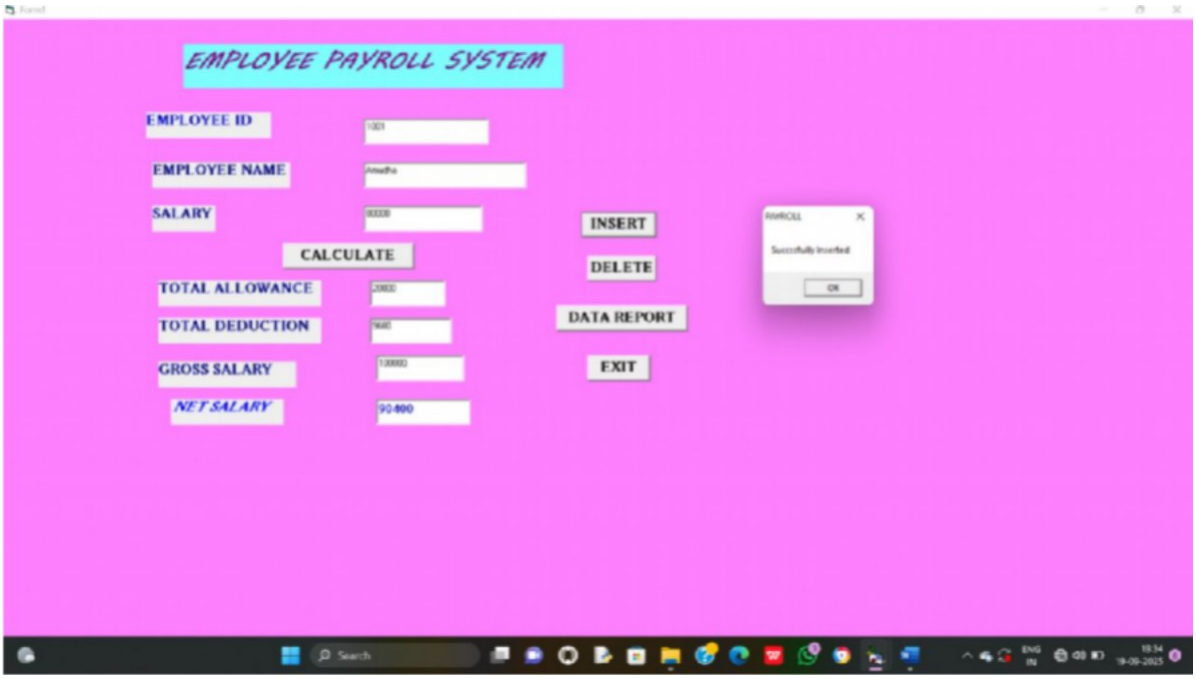
    Try
      cn.Open()
      Dim cmd As New OleDbCommand("DELETE FROM emp WHERE empid = " & eid, cn)
      Dim rowsAffected As Integer = cmd.ExecuteNonQuery()

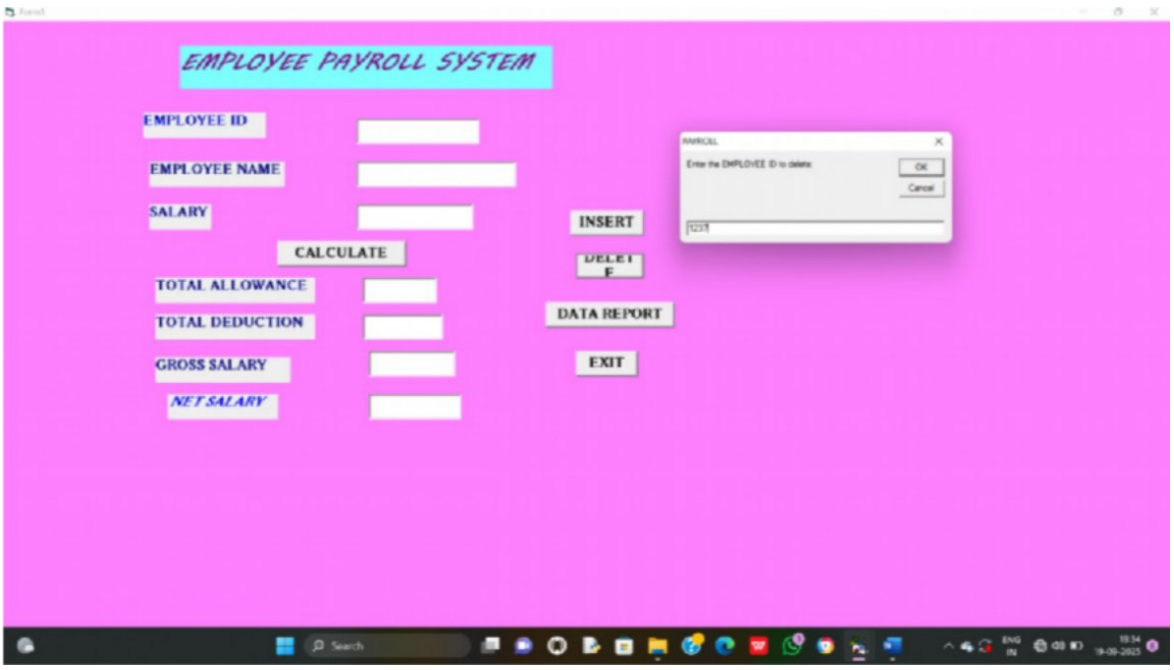
      If rowsAffected > 0 Then
        MsgBox("Record with ID " & eid & " deleted successfully!")
      Else
        MsgBox("No record found with ID: " & eid)
      End If
    Catch ex As Exception
      MsgBox(ex.Message)
    Finally
      cn.Close()
    End Try
  End Sub

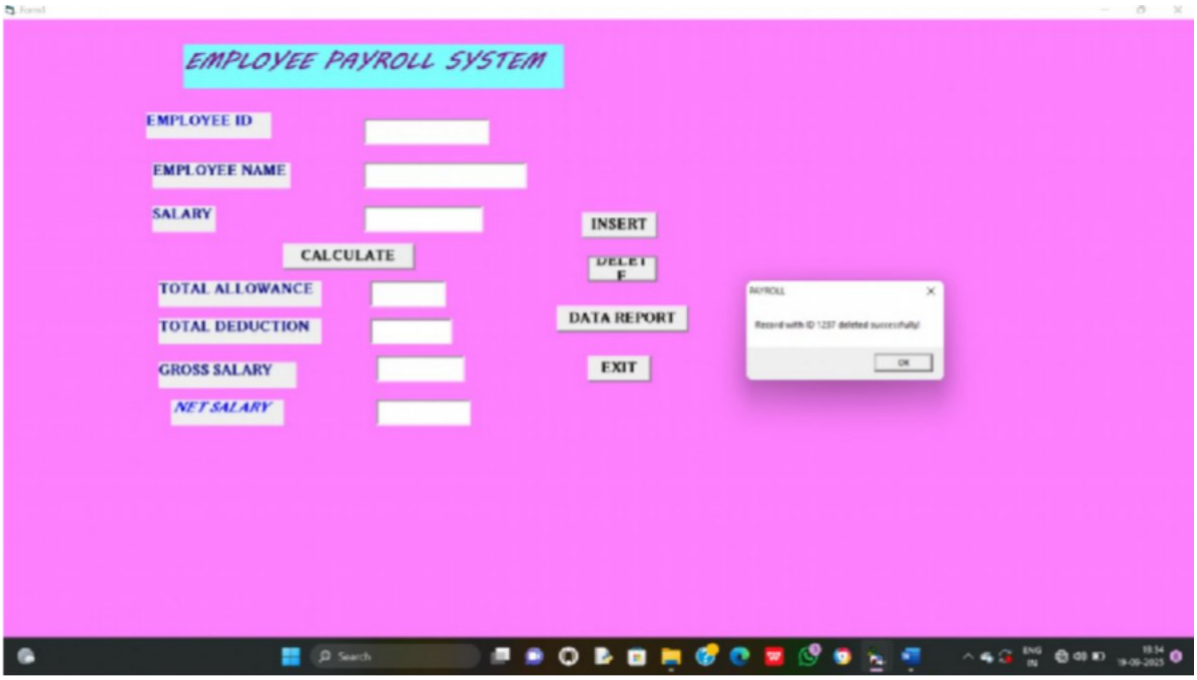
  ' --- VIEW ---
  Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
    Try
      Dim da As New OleDbDataAdapter("SELECT * FROM emp", cn)
      Dim dt As New DataTable()
      da.Fill(dt)
      DataGridView1.DataSource = dt
    Catch ex As Exception
      MsgBox(ex.Message)
    End Try
  End Sub

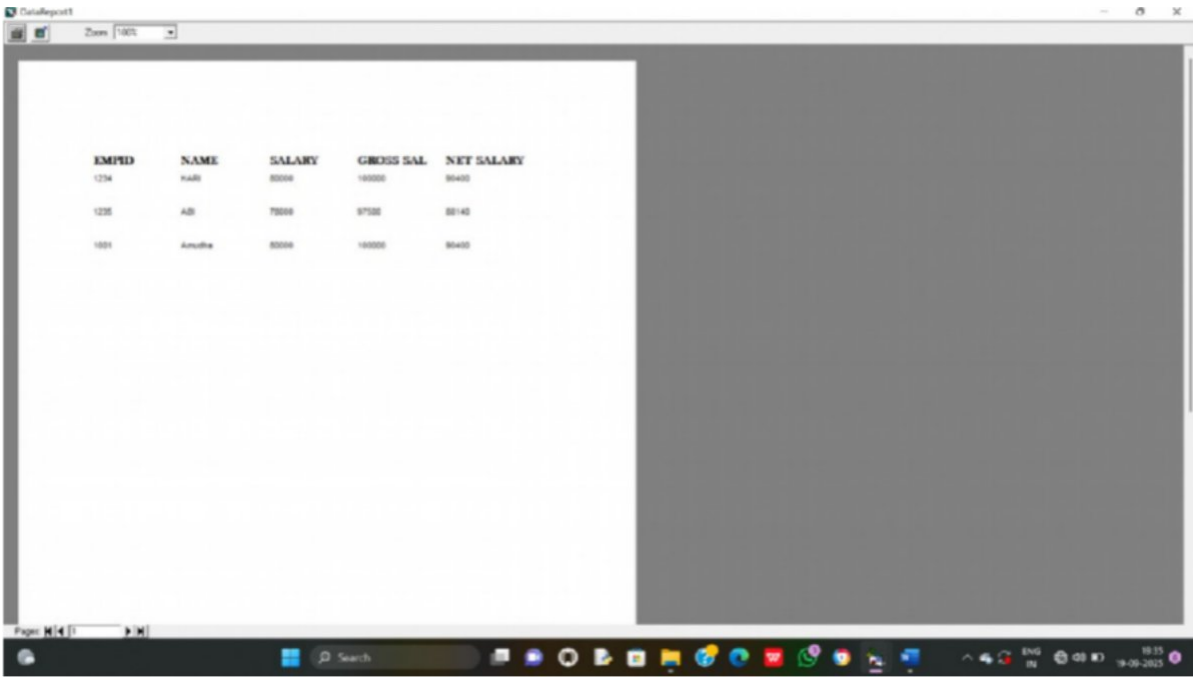
  ' --- EXIT ---
  Private Sub Button5_Click(sender As Object, e As EventArgs) Handles Button5.Click
    Application.Exit()
  End Sub
End Class
```

Output:


EMPLOYEE PAYROLL SYSTEM
EMPLOYEE ID
EMPLOYEE NAME
SALARY
CALCULATE
TOTAL ALLOWANCE
TOTAL DEDUCTION
GROSS SALARY
NET SALARY
INSERT
DELETE
DATA REPORT
EXIT
PAYROLL
Succesfully Inserted
OK


EMPLOYEE PAYROLL SYSTEM
EMPLOYEE ID
EMPLOYEE NAME
SALARY
CALCULATE
TOTAL ALLOWANCE
TOTAL DEDUCTION
GROSS SALARY
NET SALARY
INSERT
DELETE
DATA REPORT
EXIT
PAYROLL
Enter the EMPLOYEE ID to delete
OK
Cancel


EMPLOYEE PAYROLL SYSTEM
EMPLOYEE ID
EMPLOYEE NAME
SALARY
CALCULATE
TOTAL ALLOWANCE
TOTAL DEDUCTION
GROSS SALARY
NET SALARY
INSERT
DELET E
DATA REPORT
EXIT
PAYROLL
Record with ID 1237 deleted successfully!
OK


DataReport1
Zoom
EMPID
NAME
SALARY
GROSS SAL
NET SALARY
1234
KARI
80000
160000
80400
1235
ABI
78000
97500
88140
1001
Amutha
80000
160000
80400
Pages